



PROJECT ABSTRACT FORM

Batch:2078

Semester: Seventh (7th)

Year: Fourth (4th)

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Project Title: Quiz Battleground

Front-end Tools: HTML, CSS, JavaScript, React.js

Back-end Tools: Node.js + Express, MongoDB

ABSTRACT

This project presents Quiz Battleground, an interactive quiz platform focused on improving online assessments through competitive gameplay while maintaining strict academic integrity. Participants enter private quiz rooms via cryptographically secured room codes and compete in rounds based on elimination gameplay. To maintain access control and fairness, the system employs a strong random room code generator coupled with the Fisher-Yates Shuffle algorithm for unbiased question order randomization. As with any online quiz, dishonest practices create the first major challenge. This specific challenge is addressed through a multi-layered cheat detection framework utilizing client-side copy-paste detection, client-side tab-switch event monitoring, and behavior flagging on the server side through timer-based checking. All these methods aim to expose and contain attempts to manipulate the system without the need for proactive automated supervision. For both players and hosts, the interface guarantees a fluid and captivating experience, thanks to the sophisticated web-based application which incorporates tools for real-time communication.

Keywords: Real-Time Quiz, Cryptography Random Code, Fisher-Yates Shuffle, Cheat Detection, Tab Switch Monitoring, Behavioral Flagging

Student Signature's

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